

Ralph Angelo Furigay

+63-927-782-4321 | f.ralphangelo13@gmail.com | linkedin.com/in/ralphangelofurigay
github.com/Rafu-00 | huggingface.co/rafurafu

EDUCATION

De La Salle University

*Bachelor of Science (Honors) in Computer Science and
Master of Science in Computer Science*

Manila, PH

October 2020 – February 2026

EXPERIENCE

Data Engineer Intern

May 2023 – July 2023

Anino Inc.

Remote

- **Migrated legacy CRON-based ingestion scripts to Apache Airflow**, implementing automatic retries and alerting to resolve persistent data loss caused by schedule failures.
- **Managed and maintained end-to-end pipelines between Firebase and BigQuery**, ensuring the high-availability and accuracy of user statistics and application event data.
- **Partnered with different departments within the company** to audit and refactor data scripts for financial reporting and user statistics, correcting calculation logic to ensure high data integrity.
- Developed and optimized ETL pipelines to **automate the delivery of critical business metrics**, reducing manual reporting overhead and improving data accessibility for stakeholders.
- **Collaborated cross-functionally with front-end and back-end developers** to support the continuous evolution of an in-house CRM, ensuring database schema alignment with new feature releases.

PROJECTS

Filipino Tokenization | *Python, PyTorch, Transformers, PEFT, Tokenizers*

May 2025 – December 2025

- Initiated research as part of a select cohort **sponsored by Google Asia Pacific**, utilizing Google Cloud Platform (GCP) credits to advance the "Development of a Filipino Tokenizer" project.
- Released language models pretrained in Filipino texts and equipped with custom-trained Filipino tokenizers.
- **Established baseline Filipino tokenization benchmarks** across different tokenization algorithms on different large language models.
- Designed and implemented Filipino byte-level tokenizers from scratch with proper Filipino text representations and forms in mind.
- **Developed a comprehensive evaluation framework** combining Morphology-focused test datasets (novel contribution) and Downstream NLP task evaluations.
- Curated new Filipino language datasets to support low-resource NLP research.
- Analyzed the impact of tokenization strategies on morphological segmentation and downstream model performance.

Coffee Bean Roast Detection | *C++, Arduino, Tensorflow*

January 2025 – April 2025

- **Quantized a Convolutional Neural Network (CNN)** using TensorFlow Lite Micro, **reducing model size** to fit within the ESP32's limited flash memory.
- **Optimized inference latency** by implementing a custom mobile-friendly architecture, achieving real-time roast classification with a memory footprint under 250 KB.
- Managed Heap/Stack allocation strategically to prevent stack overflow during tensor arena initialization, ensuring stable runtime performance during continuous monitoring.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS

Data & Cloud: Google Cloud Platform GPU instances, Apache Airflow, BigQuery, Firebase

Developer Tools: Git, Docker, Google Cloud Platform, VS Code, JupyterLab, Conda, pip

Libraries: pandas, NumPy, Matplotlib, PyTorch, Hugging Face Transformers, scikit-learn